stretch out the napkins in every direction while on some kind of infinite treadmill running at an extreme speed over the course of billions of years, and hopefully it'll make sense then. Meanwhile, we should get back to what this all means for the future of existence. Because it really isn't good.

THE SLOW FADE TO BLACK

The assertion that "dark energy ruins everything" is not an overstatement. A universe whose expansion is accelerating is, paradoxically, one in which the influence exerted by the things in it is shrinking. Distant galaxies being dragged out of the Hubble radius by cosmic expansion will become lost to us. Galaxies whose distant past we can see now will slowly fade into darkness like ancient decaying photographs. In our own cosmic neighborhood, after the Milky Way and Andromeda merge, our little Local Group of galaxies will become more and more isolated, surrounded by darkness and the dying primordial light. All across the cosmos, invisible to us, other groups and clusters of galaxies will merge to form giant elliptical clumps of stars, burning brightly in the initial violence of the collisions but fading eventually to embers, whose glow will never reach beyond their own pool of expanding, emptying space.

Eventually, each new, dying supergalaxy will be utterly alone. Nothing will again approach to bring in a fresh supply of gas to fuel new stars. The stars already shining will burn out, exploding as supernovae or, more often, sloughing off outer layers to become slow-burning relics, gradually cooling for billions or trillions of years. Black holes will grow, for a time. Some will engulf galaxies' worth of dead stellar remnants; some will stall in their growth, with no new matter approaching close enough to be consumed.

When the stars have all faded to darkness, the ultimate decay sets in.

Black holes begin to evaporate.

It was originally thought that black holes were eternal—capable of growing by consuming other matter but incapable of ever losing any mass. It makes sense that something defined by the fact that not even light can escape it would be a one-way, bottomless pit. But Stephen Hawking calculated in the 1970s that quantum effects on a black hole's horizon cause it to glow, faintly. The glow carries away energy—or, equivalently, mass—and the black hole shrinks. This process goes slowly at first, and then faster and brighter and hotter until a final explosion and disappearance at the end.